

E.G.S. PILLAY ENGINEERING COLLEGE, (Autonomous)
NAGAPATTINAM – 611 002.

(Affiliated to Anna University, Chennai | Accredited by NAAC with 'A++' Grade
Accredited by NBA | Approved by AICTE, New Delhi)



REGULATIONS - R2023

B.E. / B.Tech. – FIFTH SEMESTER CURRICULUM

Fifth Semester											
<u>S.No</u>	COURSE CODE	COURSE NAME	CATEGORY	L	T	P	C	MAX. MARKS			
								CA	ES	TOTAL	
Theory Courses											
1	2302ME501	Design of Machine Elements	PCC	3	1	0	4	40	60	100	
2	2302ME502	Engineering Materials and Metallurgy	PCC	3	0	0	3	40	60	100	
3	2302ME503	Kinematics of Machines	PCC	2	1	0	3	40	60	100	
4		Professional Elective I	PEC	3	0	0	3	40	60	100	
5		Professional Elective II	PEC	3	0	0	3	40	60	100	
6		Open Elective – I	OEC	3	0	0	3	40	60	100	
Laboratory Courses											
7	2302ME551	Computer Aided Design Laboratory	PCC	0	0	2	1	60	40	100	
8	2302ME552	Material Testing and Characterization Laboratory	PCC	0	0	4	2	60	40	100	
9	2304GE501	Professional Development Course -III	EEC	0	0	2	1	100	-	100	
10		Life skill course - V	MC	0	0	0	0	100	-	100	
			TOTAL	17	2	8	23	560	440	1000	

L – Lecture | T – Tutorial | P – Practical | CA – Continuous Assessment | ES – End Semester

2302ME501	DESIGN OF MACHINE ELEMENTS				L	T	P	C			
					3	1	0	4			
PREREQUISITE											
1. Engineering mechanics											
2. Strength of Materials											
COURSE OBJECTIVES:											
To learn the various steps involved in the Design Process.											
To Learn designing shafts and couplings for various applications.											
To Learn the design of temporary and permanent Joints.											
To Learn designing helical, leaf springs, flywheels, connecting rods and crank shafts for various applications.											
To Learn designing and select sliding and rolling contact bearings, seals and gaskets. (Use of PSG Design Data book is permitted)											
MODULE I FUNDAMENTAL CONCEPTS IN DESIGN 12 Hours											
Introduction to the design process - factors influencing machine design, selection of materials based on mechanical properties- Direct, Bending and torsional loading- Modes of failure - Factor of safety –Principal stresses – Design of straight and curved beams – Crane hook and ‘C’ frame- Theories of failure – stress concentration – Design for variable loading - Soderberg, Goodman, Gerber methods and combined stresses											
MODULE II DESIGN OF SHAFTS AND COUPLINGS 12 Hours											
Design of solid and hollow shafts based on strength, rigidity and critical speed – Keys and splines – Rigid and flexible couplings.											
MODULE III DESIGN OF TEMPORARY AND PERMANENT JOINTS 12 Hours											
Threaded fasteners - Bolted joints including stresses due to static loading, eccentrically loading – Welded joints -Butt, Fillet and parallel transverse fillet welds – welded joints subjected to bending, torsional and eccentric loads											
MODULE IV DESIGN OF ENERGY STORING ELEMENTS AND ENGINE COMPONENTS 12 Hours											
Types of springs, design of helical springs –surge in springs, concentric springs, Design of leaf springs - rubber springs - Flywheels considering stresses in rims and arms for engines											
MODULE V DESIGN OF BEARINGS AND MISCELLANEOUS ELEMENTS 12 Hours											
Types of bearing- Hydrodynamic journal bearings, Sommerfeld Number- Ball bearing and rolling contact bearings - Selection of Rolling Contact bearings											
TOTAL: 60 HOURS											
COURSE OUTCOMES:											
On the successful completion of the course, students will be able to											
CO1:	Explain the design machine members subjected to static and variable loads.										
CO2:	Apply the concepts design to shafts, key and couplings.										
CO3:	Apply the concepts of design to bolted, Knuckle, Cotter, riveted and welded joints.										
CO4:	Apply the concept of design helical, leaf springs, flywheels, connecting rods and crank shafts.										
CO5:	Apply the concepts of design and select sliding and rolling contact bearings, seals and gaskets.										
COs Vs POs MAPPING:											
Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	3					1	1		2
CO2	2	2	3					1	1		2
CO3	2	2	3					1	1		2

CO4	2	2	3					1	1		2
CO5	2	2	3					1	1		2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1			3
CO2			3
CO3			3
CO4			3
CO5			3

REFERENCES:

1. Bhandari V B, “Design of Machine Elements”, 4th Edition , Tata McGraw-Hill Book Co, 2016
2. Joseph Shigley, Richard G. Budynas and J. Keith Nisbett “Mechanical Engineering Design”, 10th Edition, Tata McGraw-Hill , 2015.
3. Ansel C Ugural, “Mechanical Design – An Integral Approach”, 1st Edition, Tata McGraw-Hill Book Co, 2004.
4. Merhyle Franklin Spotts, Terry E. Shoup, and Lee EmreyHornberger, “Design of Machine Elements” 8th Edition, Printice Hall, 2004.
5. Robert C. Juvinall and Kurt M. Marshek, “Fundamentals of Machine component Design”,6th Edition, Wiley, 2017.
6. Sundararamamoorthy T. V. and Shanmugam .N, “Machine Design”, Anuradha Publications, Chennai, 2003.
7. Design of Machine Elements | SI Edition | Eighth Edition | By Pearson by M. F. Spotts, Terry E. Shoup, et al. | 25 March 2019

2302ME502	ENGINEERING MATERIALS AND METALLURGY	L	T	P	C
		3	0	0	3
PREREQUISITE					
1. Basic Physics 2. Basic Chemistry					
COURSE OBJECTIVES:					
To learn the constructing the phase diagram and using of iron-iron carbide phase diagram for microstructure formation.					
2 To learn selecting and applying various heat treatment processes and its microstructure formation.					
3 To illustrate the different types of ferrous and non-ferrous alloys and their uses in engineering field.					
4 To illustrate the different polymer, ceramics and composites and their uses in engineering field.					
5 To learn the various testing procedures and failure mechanism in engineering field.					
MODULE I					
CONSTITUTION OF ALLOYS AND PHASE DIAGRAMS					9 Hours
Constitution of alloys – Solid solutions, substitutional and interstitial – phase diagrams, Isomorphous, eutectic, eutectoid, peritectic, and peritectoid reactions, Iron – Iron carbide equilibrium diagram. Classification of steel and cast-Iron microstructure, properties and application.					
MODULE II					
HEAT TREATMENT					9 Hours
Definition – Full annealing, stress relief, recrystallisation and spheroidising –normalizing, hardening and tempering of steel. Isothermal transformation diagrams – cooling curves superimposed on I.T. diagram – continuous cooling Transformation (CCT) diagram – Austempering, Martempering – Hardenability, Jominy end quench test -case hardening, carburizing, Nitriding, cyaniding, carbonitriding – Flame and Induction hardening – Vacuum and Plasma hardening – Thermo-mechanical treatments- elementary ideas on sintering.					
MODULE III					
FERROUS AND NON-FERROUS METALS					9 Hours
Effect of alloying additions on steel (Mn, Si, Cr, Mo, Ni, V,Ti& W) – stainless and tool steels – HSLA – Maraging steels – Grey, white, malleable, spheroidal – alloy cast irons, Copper and its alloys – Brass, Bronze and Cupronickel – Aluminium and its alloys; Al-Cu – precipitation strengthening treatment – Titanium alloys, Mg-alloys, Ni-based super alloys – shape memory alloys- Properties and Applications overview of materials standards					
MODULE IV					
NON-METALLIC MATERIALS					9 Hours
Polymers – types of polymers, commodity and engineering polymers – Properties and applications of PE, PP, PS, PVC, PMMA, PET, PC, PA, ABS, PAI, PPO, PPS, PEEK, PTFE, Thermoset polymers – Urea and Phenol formaldehydes –Nylon, Engineering Ceramics – Properties and applications of Al2O3, SiC, Si3N4, PSZ and SIALON – intermetallics- Composites- Matrix and reinforcement Materials applications of Composites – Nano composites.					
MODULE V					
MECHANICAL PROPERTIES AND DEFORMATION MECHANISMS					9 Hours
Mechanisms of plastic deformation, slip and twinning – Types of fracture – fracture mechanics- Griffith's theory- Testing of materials under tension, compression and shear loads – Hardness tests (Brinell, Vickers and Rockwell), Micro and nano-hardness tests, Impact test Izod and charpy, fatigue and creep failure mechanisms.					
TOTAL: 45 HOURS					
COURSE OUTCOMES:					
On the successful completion of the course, students will be able to					
CO1:	Explain alloys and phase diagram, Iron-Iron carbon diagram and steel classification.				
CO2:	Explain isothermal transformation, continuous cooling diagrams and different heat treatment				

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	1	3	2							2
CO2	3	1	3	1		2		1			2
CO3	3	1	3								2
CO4	3	1	3				2				2
CO5	3	1	3	2							2

COs	PSO1	PSO2	PSO3
CO1	2		1
CO2	2		1
CO3	2		1
CO4	2		1
CO5	2		1

1. Kenneth G.Budinski and Michael K. Budinski, “Engineering Materials”, Prentice Hall of India Private Limited, 9th edition ,2018.
2. Sydney H.Avner, “Introduction to Physical Metallurgy”, McGraw Hill Book Company, 1994
3. Alavudeen, N. Venkateshwaran, and J. T.Winowlin Jappes, A Textbook of Engineering Materials and Metallurgy, Laxmi Publications, 2006.
4. Amandeep Singh Wadhwa, and Harvinder Singh Dhaliwal, A Textbook of Engineering Material and Metallurgy, University Sciences Press, 2008.
5. G.S. Upadhyay and Anish Upadhyay, “Materials Science and Engineering”, Viva Books Pvt.Ltd, New Delhi, 2020.
6. Raghavan.V, “Materials Science and Engineering”, Prentice Hall of India Pvt.Ltd. 6th edition, 2019.
7. Williams D Callister, “Material Science and Engineering” Wiley India Pvt Ltd, 2nd edition Re print 2019.

2302ME503	KINEMATICS OF MACHINES	L	T	P	C
		2	1	0	3
PREREQUISITE					
1. Engineering Mechanics 2. Mathematics 3. Engineering Graphics					
COURSE OBJECTIVES:					
1. To understand the basic components and layout of linkages in the assembly of a system/ machine. 2. To understand the principles in analyzing the assembly with respect to the displacement, velocity, and acceleration at any point in a link of a mechanism. 3. To understand the motion resulting from a specified set of linkages, design few linkage mechanisms and cam mechanisms for specified output motions. 4. To understand the basic concepts of toothed gearing and kinematics of gear trains and the effects of friction in motion transmission and in machine components.					
MODULE I FUNDAMENTALS OF MECHANISMS					
					9 Hours
Basic Terminology - Kinematic link, Pair, joints, Structure, Machine, Degree of freedom, Grubler & Kutzbach Criterion - Inversions of four bar mechanism, Mechanical advantage - Transmission Angle, Inversion of single slider and double slider crank mechanisms. Common Mechanisms - Straight line mechanism, Dwell mechanism.					
MODULE II KINEMATIC ANALYSIS OF MECHANISMS					9 Hours
Relative velocity of kinematic link, Rubbing Velocity of kinematic pair, Construction of velocity and acceleration diagram by graphical method (Relative Velocity Method), Four bar mechanism, slider crank mechanisms and complex mechanism.					
MODULE III CAM AND FOLLOWER MECHANISMS					9 Hours
Introduction - Terminology, Classifications, Types of follower motion - Uniform Velocity Motion, Simple Harmonic Motion, Uniform Acceleration and Retardation Motion and Cycloidal Motion- Construction of cam profile - Knife edge follower, Roller and flat faced follower.					
MODULE IV GEAR AND GEAR TRAIN					9 Hours
Gears - Terminology, Law of gearing, Length of path of contact, Length of arc of contact, contact ratio- Interference and undercutting. Gear trains- Speed ratio, train value. Simple gear train, compound gear train, Epicyclic gear train- speed calculation by tabular method.					
MODULE V FRICTION DRIVES					9 Hours
Introduction-Friction clutch, types -single plate, Multi plate and cone clutch. Flat Belt Drives Velocity, slip, creep and Centrifugal effect of belt, length of open and cross belt drives, Maximum power transmitted, ratio of driving tension in flat belt drives - V Belt drives.					
TOTAL: 45 HOURS					
COURSE OUTCOMES:					
On the successful completion of the course, students will be able to					
CO1:	Differentiate the basic machine mechanisms.				
CO2:	Calculate velocity and acceleration of machine mechanisms.				
CO3:	Construct the cam profile for different types of follower motion.				
CO4:	Describe the kinematic terminologies of spur gear and calculate speed ratio of various types of gear train				

2302ME551	COMPUTER AIDED DESIGN LABORATORY				L	T	P	C			
					0	0	2	1			
PREREQUISITE											
1. Engineering Graphics 2. Engineering Drawing											
COURSE OBJECTIVES											
1. Introduce students to 3D modelling software 2. Develop skills in creating 3D part models of mechanical components. 3. Teach assembly modelling techniques for machine elements. 4. Enhance understanding of real-world mechanical systems through 3D visualization. 5. Prepare students for industry-standard design practices and simulation tools.											
3D GEOMETRIC MODELLING LIST OF EXPERIMENTS:											
1. Introduction of 3D Modelling software 2. Create the part model of mechanical components											
Creation of 3D assembly model of following machine elements using 3D Modelling software											
1. Flange Coupling 2. Plummer Block 3. Screw Jack 4. Universal Joint 5. Machine Vice 6. Stuffing box 7. Safety Valves 8. Non-return valves 9. Connecting rod 10. Piston 11. Crankshaft											
TOTAL: 30 HOURS											
COURSE OUTCOMES:											
After completing this course, students will be able to:											
CO1:	Understand the interface and tools of 3D modelling software.										
CO2:	Create 3D part models of mechanical components with proper dimensions and constraints.										
CO3:	Assemble machine components using mating conditions (constraints, joints, etc.).										
CO4:	Analyze the functionality of mechanical assemblies through virtual simulation.										
CO5:	Apply industry best practices in 3D modelling for real-world applications.										
COs Vs POs MAPPING:											
Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3			3							
CO2	3	3	3	3							3
CO3	3	3	3	3	3						3
CO4	3	3	3	3	3						3
CO5			3	3	3						3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1		2	3
CO2		2	3
CO3		2	3
CO4		2	3
CO5		2	3

REFERENCE

1. Automation, Production system & Computer Integrated manufacturing, M. P. Groover Person India, 2007 2nd edition.
2. Principles of Computer Integrated Manufacturing, S. Kant Vajpayee, Prentice Hall India.
3. Ibrahim Zeid, CAD/ CAM Theory and Practice, McGraw Hill, 2007
4. Mikell P. Groover and Emory W. Zimmer, CAD/ CAM – Computer aided design and manufacturing, Pearson Education, 1987
5. T. R. Chandrupatla and A. D. Belagundu, Introduction to Finite Elements in Engineering, Pearson Education, 2012

2302ME552	MATERIAL TESTING AND CHARACTERIZATION LABORATORY	L	T	P	C						
		0	0	4	2						
PREREQUISITE											
1. Engineering Materials 2. Physics and Chemistry 3. Metrology and Measurement Techniques											
COURSE OBJECTIVES											
1. To understand and analyze the wear behavior of different materials under dry and lubricated conditions 2. To investigate microstructural features using optical microscopy and hardness measurements. 3. To develop skills in the preparation of specimens for metallurgical examination. 4. To explore fatigue and delamination behavior in various materials. 5. To program and operate Wire Electrical Discharge Machining (WEDM) for precision machining applications.											
LIST OF EXPERIMENTS											
1. Study of Dry Sliding wear of a various materials 2. Study of wear testing of various materials with lubricant recirculation 3. Study of wear testing with Pin heating of various materials 4. Fatigue delamination of materials 5. Micro hardness testing on alloy and composite 6. Optical microscopic study using video microscope 7. Specimen preparation for micro structure examination 8. Programming of WEDM process parameter 9. Machining of various model using WEDM.											
TOTAL: 30 HOURS											
COURSE OUTCOMES:											
On the successful completion of the course, students will be able to											
CO1:	Analyze wear characteristics of materials under dry, lubricated, and thermal conditions.										
CO2:	Perform fatigue and delamination tests and interpret failure mechanisms.										
CO3:	Conduct micro-hardness testing and prepare specimens for microstructural examination.										
CO4:	Use optical microscopy to identify and interpret microstructures in alloys and composites.										
CO5:	Program WEDM parameters and machine complex shapes with precision.										
COs Vs POs MAPPING:											
Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	2	2					2	2
CO2	3	2	2	2						2	2
CO3	3		2								1
CO4	3		2		2						1
CO5	3	2	2	2	3						2
COs Vs PSOs MAPPING:											
Cos	PSO1	PSO2	PSO3								
CO1	3		1								

	CO2	3		1
	CO3	3		1
	CO4	3		1
	CO5	3		1
Reference 1. Callister, W.D., & Rethwisch, D.G, Materials Science and Engineering: An Introduction (10th ed.) 2020 2. Kalpakjian, S., & Schmid, S.R., Manufacturing Engineering and Technology (7th ed.). Pearson Education., 2014 3. ASM International., ASM Handbook Volume 8: Mechanical Testing and Evaluation. ASM International., 2002 4. Serope Kalpakjian & Steven R. Schmid, Materials and Processes in Manufacturing (11th ed.), 2016				